

# AI-driven digital twins for manufacturing process modeling and optimization.

Manufacturing × Digital twin and simulation × Machine learning · OS097 v1 · refreshed 2026-08-18

| ATTRACTIVENESS                   | RIGHT TO WIN                         | COMPETITION                              | SERVICEABLE MARKET                           | HORIZON                                             | PORTFOLIO               |
|----------------------------------|--------------------------------------|------------------------------------------|----------------------------------------------|-----------------------------------------------------|-------------------------|
| <b>44</b><br>is the world moving | <b>74</b><br>can we play, can we win | <b>MEDIUM</b><br>0 named in the evidence | <b>€309m</b><br>observed evidence · per year | <b>LATER</b><br>research and standards<br>activi... | <b>L1</b><br>17 signals |

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

## The opportunity

AI-driven digital twins for manufacturing process modeling and optimization let manufacturers monitor, predict, and optimize production in real time, improving quality and efficiency. This opportunity targets manufacturing leaders facing quality variations and costly data acquisition, and it exists now because AI and digital twin research has matured enough to bridge physical and virtual domains.

## Market opportunity

| Method                                             | Total (TAM)                      | Serviceable (SAM)                | Obtainable (SOM)                | Confidence |
|----------------------------------------------------|----------------------------------|----------------------------------|---------------------------------|------------|
| Bottom-up: enterprises x adoption x contract value | <b>€431m</b><br>€77.6m – €2.3bn  | <b>€309m</b><br>€55.7m – €1.7bn  | <b>€14.9m</b><br>€2.7m – €79.8m | observed   |
| Observed: tendered contract value, annualised      | <b>€20.9m</b><br>€20.9m – €56.6m | <b>€20.9m</b><br>€20.9m – €56.6m | <b>€1.0m</b><br>€1.0m – €2.7m   | observed   |

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

### Bottom-up: enterprises x adoption x contract value

| Factor                                             | Value                                                              | Basis      | Source                                        | As of                  |
|----------------------------------------------------|--------------------------------------------------------------------|------------|-----------------------------------------------|------------------------|
| Enterprises in Manufacturing (EU27_2020)           | 238,122 enterprises                                                | observed   | Eurostat · sbs_sc_ovw                         | 2024                   |
| Enterprises adopting Machine learning              | 3.3 % of enterprises (10+ employees)                               | observed   | Eurostat · isoc_eb_ain2 · E_AI_TML            | 2025                   |
| Annual value of one engagement                     | €337k per enterprise per year                                      | observed   | TED (Tenders Electronic Daily) · ted          | 2026-05-19..2026-08-05 |
| Size-weighted buyer base                           | 39,220 enterprise-equivalents at large-enterprise engagement value | assumption | config/sizing.yaml · size_class_value_weights | size-2026-08-a         |
| Assumed obtainable share of the serviceable market | 4.8 % of SAM                                                       | assumption | config/sizing.yaml · obtainable_share         | size-2026-08-a         |

**Read this before quoting the number.** Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 2 of 3 declared geographies are outside the European reference data (CN, US); the estimate covers the European footprint only.

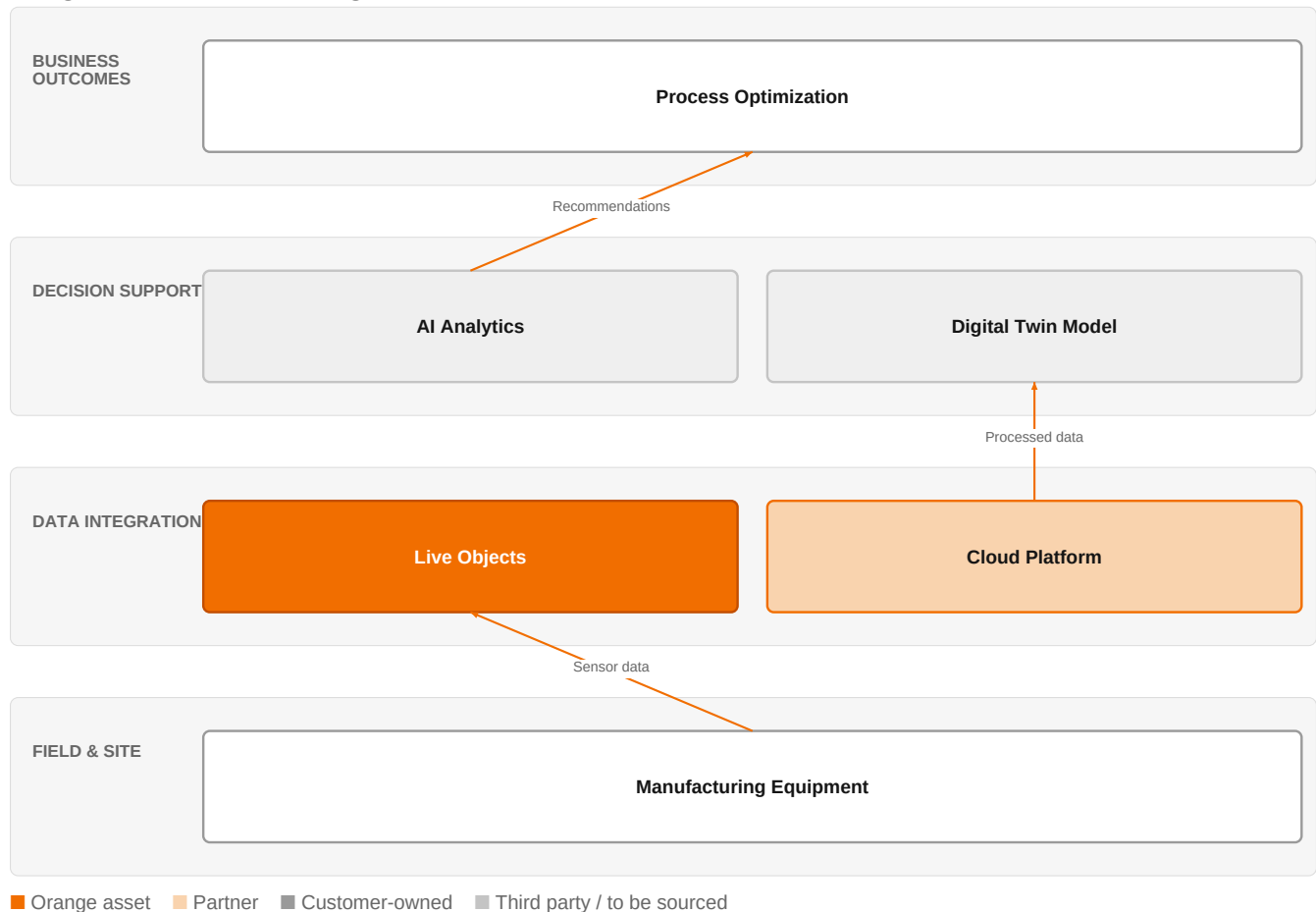
### Observed: tendered contract value, annualised

| Factor                                             | Value           | Basis      | Source                                | As of                  |
|----------------------------------------------------|-----------------|------------|---------------------------------------|------------------------|
| Tendered contract value in matching CPV categories | €20.9m per year | observed   | TED (Tenders Electronic Daily) · ted  | 2026-05-19..2026-08-05 |
| Assumed obtainable share of the serviceable market | 4.8 % of SAM    | assumption | config/sizing.yaml · obtainable_share | size-2026-08-a         |

**Read this before quoting the number.** Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 2 of 3 declared geographies are outside the European reference data (CN, US); the estimate covers the European footprint only.

## The solution, in one picture

### AI Digital Twin for Manufacturing



This diagram shows how Orange integrates IoT data from manufacturing equipment into a cloud-based digital twin that drives process optimization.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

## What has changed

Digital twins integrated with AI are emerging as transformative tools in smart manufacturing, bridging physical and virtual domains to enable real-time monitoring, predictive analytics, and autonomous decision-making. AI enhances the ability to acquire process knowledge, optimize scheduling, and autonomously control system variables. Recent research demonstrates digital twin frameworks for multi-objective optimization in semiconductor manufacturing and AI-augmented additive manufacturing, embedding AI as a central framework for intelligent, adaptive, and autonomous systems. These advances address the high cost of data acquisition in industrial environments by using AI to model process systems and improve quality through model-based control and optimization.

Sources: SIG-99692E33DB88 openalex.org 2025-12-24; SIG-4C032C4148C6 openalex.org 2026-01-01; SIG-9CFA159A98D6 openalex.org 2026-01-06; SIG-AE9A0D62F13D openalex.org 2026-04-01

## Who buys, and why now

The manufacturing operations director or VP of manufacturing signs, while the process engineer feels the pain of product quality variations and costly data acquisition. The trigger is a quality issue or a new production line where existing models fail to predict performance, forcing a search for better process control and optimization.

## Why it is hot now — every claim bound to a dated source

- Digital Twins integrated with AI bridge the physical and virtual domains, enabling real-time monitoring, predictive analytics, and autonomous decision-making. [SIG-99692E33DB88]
- AI enhances the ability to acquire process knowledge, optimize scheduling, and autonomously control system variables. [SIG-99692E33DB88]

- Digital twins integrated with AI enable real-time monitoring, predictive analytics, and autonomous decision-making in manufacturing. [SIG-99692E33DB88]
- A digital twin framework with deep feature extraction and Gaussian process enables multi-objective optimization in semiconductor manufacturing. [SIG-4C032C4148C6]
- Digital twin models for manufacturing process systems can improve quality through model-based process control and optimization. [SIG-AE9A0D62F13D]
- AI-augmented additive manufacturing embeds AI as a central framework driving intelligent, adaptive, and autonomous additive manufacturing systems. [SIG-9CFA159A98D6]
- A digital twin hierarchy for metal additive manufacturing optimizes design and manufacturing processes, improving product strength and reducing mass. [SIG-A067578BB3A4]

## What Orange would deliver, and why it can win

### What Orange would deliver

Orange would assemble a solution using its AI/Data/Cloud experts, IoT experts, and Orange Group researchers, leveraging Live Objects for IoT data ingestion, and partnering with AWS, Google Cloud, NiCE, or NVIDIA for compute and AI platforms. The engagement would start with a discovery workshop to map the customer's process and data, then build a digital twin model using machine learning, deploy it on a cloud platform, and integrate it with the customer's existing systems for real-time monitoring and optimization.

### Why Orange can win

Orange brings certified expertise (ISO 27001, TISAX) and a track record with manufacturing references like Colgate-Palmolive, Mondelez International, and Saint-Gobain Glass. The combination of IoT expertise, cloud partnerships, and research capability allows Orange to deliver a sovereign, secure digital twin solution that hyperscalers or pure AI platforms may not offer with the same level of integration and trust.

### Named assets behind this — each individually inspectable (LK-08)

| Asset                     | Type            | Link | Why it is linked                                           | Curator     |
|---------------------------|-----------------|------|------------------------------------------------------------|-------------|
| Live Objects              | offer           | L1   | offer provides the technology in a matching domain         | unconfirmed |
| AWS                       | partner         | L2   | partner provides the required technology                   | unconfirmed |
| Google Cloud              | partner         | L2   | partner provides the required technology                   | unconfirmed |
| NiCE                      | partner         | L2   | partner provides the required technology                   | unconfirmed |
| NVIDIA                    | partner         | L2   | partner provides the required technology                   | unconfirmed |
| ISO 27001                 | certification   | SUP  | certification Orange holds is required by this vertical    | unconfirmed |
| TISAX                     | certification   | SUP  | certification Orange holds is required by this vertical    | unconfirmed |
| AI / Data / Cloud experts | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| IoT experts               | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Orange Group researchers  | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Colgate-Palmolive         | reference       | SUP  | published reference in this vertical, different use case   | unconfirmed |
| Mondelez International    | reference       | SUP  | published reference in this vertical, different use case   | unconfirmed |
| Saint-Gobain Glass        | reference       | SUP  | published reference in this vertical, different use case   | unconfirmed |
| Skytale                   | reference       | SUP  | published reference in this vertical, different use case   | unconfirmed |

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

## Competitive landscape

**MEDIUM** competitive intensity (score 7.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

| Competitor    | Category                     | Position here                                                                     | Basis      | Evidence |
|---------------|------------------------------|-----------------------------------------------------------------------------------|------------|----------|
| AWS           | Hyperscaler                  | sells Machine learning; competes in OX: Smart Industries — also an Orange partner | structural | —        |
| Google Cloud  | Hyperscaler                  | sells Machine learning — also an Orange partner                                   | structural | —        |
| Atos / Eviden | Global systems integrator    | sells Machine learning                                                            | structural | —        |
| NICE          | Customer-experience platform | sells Machine learning — also an Orange partner                                   | structural | —        |
| Databricks    | Data and AI platform         | sells Machine learning                                                            | structural | —        |
| Dataiku       | Data and AI platform         | sells Machine learning                                                            | structural | —        |
| Palantir      | Data and AI platform         | sells Machine learning; competes in OX: Smart Industries                          | structural | —        |
| Accenture     | Global systems integrator    | active in Manufacturing; competes in OX: Smart Industries                         | structural | —        |

**evidenced** means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

## Qualifying questions for the first meeting

- 1 What specific quality variations or process inefficiencies are you currently experiencing that you believe a digital twin could address?
- 2 How do you currently collect and store process data from your manufacturing lines, and what is the volume and granularity?
- 3 Have you attempted any simulation or digital twin projects before? If so, what were the outcomes and challenges?
- 4 Which production processes are your highest priority for optimization, and what are the key performance indicators you would use to measure success?
- 5 What is your organization's stance on cloud adoption and data sovereignty, and are there any regulatory constraints on where data can be processed?

## Objections you will meet

| They will say                                                    | You can answer                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| We already have a digital twin initiative with a hyperscaler.    | That's a strong starting point. Our value is in integrating that with your existing IoT and operational systems, and in bringing manufacturing-specific expertise and security certifications that ensure the solution is both effective and compliant.          |
| The cost of building and maintaining a digital twin is too high. | We understand cost is a concern. Our approach leverages existing cloud platforms and our own AI expertise to reduce development time, and we can start with a focused pilot on one process to demonstrate ROI before scaling.                                    |
| We're worried about data security and intellectual property.     | That's a valid concern. Orange holds ISO 27001 and TISAX certifications, and we can deploy on sovereign infrastructure or on-premises if needed. We also have experience with sensitive manufacturing data from our work with companies like Saint-Gobain Glass. |

## Risks and unknowns

The main risk is that digital twin technology is still evolving, with challenges like concept drift and model reusability across different configurations. The customer may be wary of the cost and complexity of implementation, and the lack of standardized best practices could slow adoption. Unknowns include the maturity of AI models for specific manufacturing processes and the willingness of customers to share sensitive production data.

## Next action, by role (FR-17)

| Role                | Do this next                                                                                                                                                                                                                                                                            |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Presales / Proposal | Prepare a bid brief for a manufacturing prospect that positions Orange's Live Objects platform, combined with AWS or Google Cloud and NVIDIA's AI capabilities, to deliver a digital twin solution, and include references to Saint-Gobain Glass and Colgate-Palmolive as proof points. |

|                        |                                                                                                                                                                                                                                                                            |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sales                  | In a conversation with a manufacturing client's operations or digital transformation lead, reference the AI-driven digital twin use case and mention Orange's expertise in IoT and cloud, citing the Live Objects platform as a foundation for real-time data integration. |
| Strategist / Innovator | Commission a deep-dive brief on AI-driven digital twins for manufacturing process modeling, leveraging Orange Group researchers and IoT experts to map the competitive landscape and identify where Live Objects can serve as the data backbone.                           |

## Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

| Signal           | Date       | Publisher    | T | Title and link                                                                                     |
|------------------|------------|--------------|---|----------------------------------------------------------------------------------------------------|
| SIG-AE9A0D62F13D | 2026-04-01 | openalex.org | 1 | Quasi-Static Hypergraph Neural Networks: A High-Performance Approach for Digital Twin Modeling ... |
| SIG-505D0AD4A3FA | 2026-02-13 | openalex.org | 1 | Digital twin model for performance analysis and manufacturing process optimization                 |
| SIG-A067578BB3A4 | 2026-02-06 | openalex.org | 1 | Optimization of the Design and Manufacturing Processes for Metal Additive Manufacturing Through... |
| SIG-9CFA159A98D6 | 2026-01-06 | openalex.org | 1 | Roadmap on Artificial IntelligenceAugmented Additive Manufacturing                                 |
| SIG-4C032C4148C6 | 2026-01-01 | openalex.org | 1 | A Digital Twin Framework With Deep Feature Extraction and Gaussian Process for Multi-Objective ... |
| SIG-8041309CEF9B | 2026-01-01 | openalex.org | 1 | Digital Twin Inference from Multi-physical Simulation Data of DED Additive Manufacturing Proces... |
| SIG-DD3447868CD6 | 2026-01-01 | openalex.org | 1 | Modeling a Process Shop Floor Digital Twin in Smart Manufacturing: Best Practices From a Case S... |
| SIG-99692E33DB88 | 2025-12-24 | openalex.org | 1 | AI-Driven Digital Twins for Manufacturing: A Review Across Hierarchical Manufacturing System Le... |
| SIG-0964410DC575 | 2025-12-17 | openalex.org | 1 | Generative and Predictive AI for digital twin systems in manufacturing                             |
| SIG-6C22EFB6A3A7 | 2025-12-17 | openalex.org | 1 | On the reusability of machine learning-based process monitoring systems for manufacturing digit... |
| SIG-1CF55C438096 | 2025-10-15 | openalex.org | 1 | Digital Twin for Automated Post-processing Chain in Additive Manufacturing                         |
| SIG-1E3EF9A3D100 | 2025-09-01 | openalex.org | 1 | An inceptive approach for designing simple digital twins and industrial metaverse process frame... |
| SIG-E4A301E44BE7 | 2025-08-26 | openalex.org | 1 | Digital twin-driven decision support approach of production process for smart manufacturing        |
| SIG-284B6A4B4E17 | 2025-08-19 | openalex.org | 1 | Design and modeling of process manufacturing-oriented digital twin system                          |
| SIG-7785660FD9FA | 2025-08-19 | openalex.org | 1 | Application of Multimodel Concept and Virtual Assistants to Build a Digital Twin for Supporting... |
| SIG-7AFE53FCEBE7 | 2026-07-20 | arxiv.org    | 3 | A Continual Validation, Updating, and Decision-Making Framework for Self-Adaptive Digital Twins... |
| SIG-9791BCD1E763 | 2026-06-16 | arxiv.org    | 3 | FacProcessTwin: An LLM-Based System for Process Twin Development                                   |

## How this brief was made

**Computed, not written.** Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

**Written, then checked.** The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive\_landscape (no claim survived evidence binding); diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

| What                        | Version                     | When                      |
|-----------------------------|-----------------------------|---------------------------|
| Scores — weight set         | w-2026-08-a                 | 2026-08-18T19:08:26+00:00 |
| Market sizing — assumptions | size-2026-08-a              | 2026-08-18T21:06:16+00:00 |
| Competitor register         | competitors-2026-08-a       | 2026-08-18T21:06:19+00:00 |
| Narrative — prompt / model  | describe-v1 / deepseek-chat | 2026-08-19T04:56:54+00:00 |
| Pipeline                    | 0.1.0                       | 2026-08-18                |

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.